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 Will be reviewed

8

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glRotatef(90.0, 0, 0, 1.0);
```

```
glScalef(-1.0, -1.0, -1.0);
```

```
glScalef(-2.0, -1.0, -1.0); *
```

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

9

Why is aspect ratio preservation important when resizing a window in OpenGL? *

- ☒ To prevent objects from appearing distorted.
- ☐ To maintain a consistent frame rate.
- ☐ To avoid aliasing artifacts.
- ☐ To improve performance.

 **Will be reviewed**

10

..... light components are view-independent *

- ☒ Ambient and Diffuse
- ☐ Ambient and Specular
- ☐ Diffuse and Specular

 **Will be reviewed**

11

The volume of a small cube changes when its vertices are rotated. *

- ☐ True
- ☒ False

 Will be reviewed

12

If you had 1,000,000 vertices and wanted to change all the vertex positions by multiplying each x-coordinate by the same vertex's y-coordinate, the fastest place to do it is: *

- ☒ in the vertex shader
- ☐ with a c++ function
- ☐ in the fragment shader

 Will be reviewed

13

What is the purpose of 'Double Buffering' in OpenGL? *

- ☒ To prevent flickering by drawing to a back buffer while the front buffer is displayed.
- ☐ To render two different scenes at the same time.
- ☐ To allow the GPU to use twice as much memory.
- ☐ To increase the resolution of the textures.

 Will be reviewed

14

Which matrix would you modify to implement a 'Zoom' effect in a camera? *

- ☐ Identity Matrix
- ☐ Projection Matrix
- ☐ Model Matrix
- ☒ View Matrix

 **Will be reviewed**

15

In Homogeneous coordinates (x, y, z, w), if $w = 1.0$, the vector represents a _____; if $w = 0.0$, it represents a _____. *

- ☐ Scale; Rotation
- ☐ Matrix; Vector
- ☒ Direction; Point
- ☐ Point; Direction

 **Will be reviewed**

16

The output of the Vertex Processor is a stream of: *

- ☐ Pixels with color information.
- ☐ Transformed vertices with updated attributes.
- ☒ Primitives that have been culled.
- ☐ 2D fragments with interpolated data.

 **Will be reviewed**

17

Which matrix is responsible for moving the "Camera" around the world? *

- ☐ Normal Matrix
- ☒ View Matrix
- ☐ Projection Matrix
- ☐ Model Matrix

 **Will be reviewed**

18

Rotating the 3D point $(-1,0,0)$ by 180 around z-axis gives what: *

- ☐ $(-1,0,0)$
- ☐ $(0,-1,0)$
- ☐ $(0,1,0)$
- ☒ $(1,0,0)$
- ☐ $(-1,-1,0)$

 **Will be reviewed**

19

Which coordinate space is the result of applying only the Model and View matrices to a local vertex? *

- ☐ View Space (or Eye Space)
- ☐ World Space
- ☐ Clip Space
- ☒ Screen Space

 **Will be reviewed**

20

Which OpenGL feature must be enabled to prevent the rendering of back faces of triangles? *

- ☐ glEnable(GL_BLEND)
- ☒ glEnable(GL_CULL_FACE)
- ☐ glEnable(GL_STENCIL_TEST)
- ☐ glEnable(GL_DEPTH_TEST)

 **Will be reviewed**

21

The output of executing the following OpenGL code segment is:

```
glBegin(GL_LINES);  
glVertex2fv(A);  
glVertex2fv(B);  
glVertex2fv(C);  
glVertex2fv(D);  
glVertex2fv(E);  
glVertex2fv(F);  
glEnd(); *
```

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

22

One important rule when you use quads is that all four corners of the quadrilateral must lie in a plane. *

☒ True

☐ False

 **Will be reviewed**

23

In OpenGL's column-major format, which indices in a 1D array of 16 elements represent the translation vector? *

☐ Indices 5, 10, 15

☐ Indices 12, 13, 14

☐ Indices 3, 7, 11

☒ Indices 0, 1, 2

 **Will be reviewed**

24

How do you calculate the inverse of a pure Translation matrix? *

- ☒ Set all diagonal elements to -1.0.
- ☐ Divide by each offset.
- ☐ Simply negate the translation offsets.
- ☐ Transpose the matrix.

 **Will be reviewed**

25

In the Graphics Pipeline, what process follows the Vertex Shader to turn vector data into fragments (pixels)? *

- ☐ Rasterization
- ☐ Alpha Blending
- ☒ Depth Testing
- ☐ Texture Mapping

 **Will be reviewed**

26

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently *

☒ True

☐ False

 **Will be reviewed**

27

Which buffer is used to ensure that objects closer to the camera are drawn over objects farther away? *

☐ Color Buffer

☐ Stencil Buffer

☐ Accumulation Buffer

☒ Depth Buffer

 **Will be reviewed**

28

OpenGL is often described as a "State Machine." What does this mean? *

☒ It maintains a set of variables (state) that determine how functions behave until they are changed.

☐ It can only render one object at a time.

☐ It automatically changes its code based on the hardware.

☐ It only works when the computer is in a specific power state.

 **Will be reviewed**

29

What happens during the "Perspective Divide" stage in the pipeline of the stages of Vertex Transformation? *

- ☐ The x, y, and z coordinates are divided by w.
- ☒ The object is clipped if it is too large.
- ☐ The colors are averaged.
- ☐ The matrix is inverted.

 **Will be reviewed**

30

A 3D point is represented in homogeneous coordinates by a 4-dimensional Vector. *

- ☒ True
- ☐ False

 **Will be reviewed**

31

Which texture wrapping option should be used to repeat an image across a surface when coordinates exceed the range [0, 1]? *

- ☐ GL_CLAMP_TO_BORDER
- ☐ GL_CLAMP_TO_EDGE
- ☒ GL_REPEAT
- ☐ GL_CLAMP

📝 Will be reviewed

32

In the Phong reflectance model, if only light component is used; a flat surface illuminated with a directional light source will have a uniform color intensity across the surface. *

☒ Ambient

☐ Diffuse

☐ Spot

☐ Specular

📝 Will be reviewed

33

Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode (GL_PROJECTION);  
glLoadIdentity();  
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);  
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glPushMatrix();  
glScalef(-0.5, -0.5,1.0);  
glRotate(180, 0, 0, 1);  
glScalef(2.0, 2.0,1.0);  
glPopMatrix();  
drawTriangle(); *
```

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

34

$\text{translate}(z,3) * \text{scale}(z, 2) = \text{scale}(z, 2) * \text{translate}(z,3) *$

☐ True

☒ False

 **Will be reviewed**

35

$\text{Translate}(1,2,3) * \text{Rotate}(90.0, 0, 0, 1.0) = \text{Rotate}(90.0, 0, 0, 1.0) * \dots\dots\dots *$

☐ $\text{Translate}(-2,1,3)$

☒ $\text{Translate}(2,-1,3)$

☐ $\text{Translate}(1,3,-2)$

☐ $\text{Translate}(1,2,3)$

 **Will be reviewed**

36

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glScalef (-2.0,1.5, -3.0); *
```

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

37

What is the primary nature of OpenGL? *

☐ A standalone programming language.

☐ A graphics editing software.

☒ A cross-language, cross-platform API for rendering 2D and 3D vector graphics.

☐ A hardware driver for GPUs.

 **Will be reviewed**

38

Which shader stage is mandatory and is responsible for calculating the screen-space coordinates of vertices? *

- ☒ Vertex Shader
- ☐ Geometry Shader
- ☐ Compute Shader
- ☐ Fragment Shader

 **Will be reviewed**

39

For a rotation matrix R and a translation matrix T , $RT = TR$, even though matrix multiplication is not generally commutative. *

- ☐ True
- ☒ False

 **Will be reviewed**

40

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(1.0f, 1.0f, 0.0f);  
glRotatef(180.0, 0, 0, 1.0);  
glBegin(GL_POINTS);  
glVertex3f(1.0, 0.0, 0.0);  
glEnd(); *
```

- ☐ (1,0,0)
- ☐ (-1,-1,0)
- ☒ (0,1,0)
- ☐ (1,-1,0)
- ☐ (-1,0,0)

 **Will be reviewed**

41

The dot product of any two perpendicular vectors equals 1 *

- ☒ True
- ☐ False

 **Will be reviewed**

42

Which lighting component in the Phong reflection model simulates the uniform light that hits all surfaces equally, even in shadows? *

- ☒ Ambient
- ☐ Specular
- ☐ Diffuse
- ☐ Emissive

 **Will be reviewed**

43

Why do we use 4x4 matrices for 3D graphics instead of 3x3 matrices? *

- ☒ To allow Translation to be represented as a matrix multiplication.
- ☐ To account for the color of the vertex.
- ☐ To increase the speed of the GPU calculations.
- ☐ To allow for 4D objects to be rendered.

 **Will be reviewed**

44

Which OpenGL function is used to specify a point in 2D space? *

- ☒ glPoint2f()
- ☐ glVertex2f()
- ☐ glDot2f()
- ☐ glPixel2i()

 **Will be reviewed**

45

The color values specified in glColor3f are typically in the range of: *

- ☒ 0 to 255
- ☐ 0.0 to 1.0
- ☐ 0 to 100
- ☐ -1.0 to 1.0

 **Will be reviewed**

46

What is the primary purpose of the glutReshapeFunc function in OpenGL? *

- ☐ To draw the scene.
- ☐ To handle keyboard input.
- ☒ To handle window resizing events.
- ☐ To initialize OpenGL.

 **Will be reviewed**

47

If you want to scale an object to be half its original size in all three dimensions, what should the first three diagonal elements of the scaling matrix be? *

- ☒ [0.5, 0.5, 0.5]
- ☐ [2, 2, 2]
- ☐ [1.5, 1.5, 1.5]
- ☐ [-0.5, -0.5, -0.5]

 Will be reviewed

48

To draw a rectangle with upper-left corner (10, 20) and lower-right corner (30, 40), how would you call `glRectf`? *

- ☐ `glRectf(20, 10, 40, 30);`
- ☒ `glRectf(10, 20, 30, 40);`
- ☐ `glRectf(40, 30, 20, 10);`
- ☐ `glRectf(30, 40, 10, 20);`

 Will be reviewed

49

The output of executing the following OpenGL code segment is:

```
glBegin(GL_QUAD_STRIP);  
glVertex2fv(A);  
glVertex2fv(B);  
glVertex2fv(C);  
glVertex2fv(D);  
glVertex2fv(E);  
glVertex2fv(F);  
glEnd(); *
```

- ☐ A
- ☐ B
- ☐ C
- ☒ D

 **Will be reviewed**

50

The OpenGL technique that takes a texture map and creates a hierarchy of coarser and coarser versions of it is called Mipmapping. *

☒ True

☐ False

 **Will be reviewed**

51

Which C-style language is used to write shaders for OpenGL? *

☐ HLSL

☐ OpenCL

☒ GLSL

☐ CUDA

 **Will be reviewed**

52

Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode (GL_PROJECTION);  
glLoadIdentity();  
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);  
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glPushMatrix();
```

```
glScalef(-1.0, 1.0,1.0);  
glRotate(180, 0, 0, 1);  
glScalef(2.0, 2.0,1.0);  
glPopMatrix();  
drawTriangle(); *
```

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

53

In the following code segment using OpenGL Shading Language, what is the value d?

```
vec4 TestVec = vec4(1.0, 2.0, 3.0, 4.0);  
vec2 d = TextVec.zx; *
```

☒ d=(3.0, 1.0)

☐ d=(4.0, 2.0)

☐ d= NULL

☐ d=(1.0, 3.0)

☐ d=(1.0, 2.0, 3.0, 4.0)

 **Will be reviewed**

54

What is the difference between orthographic and perspective projection? *

- ☐ Orthographic projection preserves parallel lines, while perspective projection does not.
- ☐ Orthographic projection uses a single vanishing point, while perspective projection uses multiple vanishing points.
- ☐ Perspective projection uses a single vanishing point, while orthographic projection uses multiple vanishing points.
- ☒ Perspective projection preserves parallel lines, while orthographic projection does not.

 **Will be reviewed**

55

If you want to rotate an object around a specific point P that is not the origin, what is the correct sequence of transformations? *

- ☒ Translate by -P, Rotate, then Translate by P.
- ☐ Scale, then Rotate, then Translate by P.
- ☐ Translate by P, Rotate, then Translate by -P.
- ☐ Rotate, then Translate to P .

 **Will be reviewed**

56

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(90.0, 0, 0, 1.0);
```

```
glPushMatrix();  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
glScalef (-1.0,0.5, -1.0);  
glPopMatrix();  
glRotatef(90.0, 0, 0, 1.0);  
glTranslatef(1.0f, -2.0f, 3.0f); *
```

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

57

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glRotatef(90.0, 0, 0, 1.0); *
```

☐ A

☐ B

☒ C

☐ D

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